

Aurobinda Majhi

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SUMMARY

Systems-focused software engineer with hands-on experience designing high-performance distributed computing frameworks, scalable search systems, and parallel processing pipelines in C, C++, and Python. Proven ability to build complex software from broadly defined problems, deliver production-quality solutions under rapidly shifting requirements, and collaborate cross-functionally in agile environments.

TECHNICAL SKILLS

Languages	C, C++, Python, JavaScript, SQL
Systems & Design	Distributed systems, object-oriented design, multithreading (POSIX), fault-tolerant architectures, scalable system design, complexity analysis
Scientific Computing	Numerical methods, HPC, parallel computing, FFTW, particle simulations
AI & ML	LLM evaluation, RLHF, prompt engineering, supervised fine-tuning, computer vision, dataset curation, multimodal AI
Libraries & Tools	Qt/QML, OpenCV, Dlib, HNSWlib, SDL2, NumPy, Pandas, Scikit-learn
Dev Tools	Git, Linux, CMake, GDB, Valgrind, Docker

PROJECTS

Distributed N-Body Cosmological Simulation & Topology Analysis *Aug 2024 – May 2025 C, Numerical Methods, HPC, FFTW, Parallel Computing* [GitHub](#)

- Designed and built a distributed particle-mesh simulation framework from scratch in C, modelling gravitational interactions across 100k+ particles using partitioned domains, FFT-based Poisson solvers, and Leapfrog integration to replicate large-scale cosmological structure formation.
- Proposed and implemented a Telegraph Equation gravitational solver as an alternative to the conventional Poisson formulation, achieving faster convergence and more accurate modelling of finite-speed gravitational propagation compared to the standard approach.
- Designed a topology-based characterisation framework for cosmic web structures, capturing morphological information inaccessible to two-point power spectrum methods and enabling richer analysis of simulation outputs.

3D Rendering Engine from Scratch *Personal Project C, SDL2, POSIX Threads* [GitHub](#)

- Architected a CPU-based 3D rendering engine from scratch in C, implementing software rasterization and recursive ray tracing pipelines without reliance on graphics APIs, demonstrating ability to design complex systems from broadly defined requirements.
- Engineered core graphics algorithms including perspective projection, scanline rasterization, Z-buffering, view-frustum clipping, Phong shading, shadow rays, and recursive reflections to produce physically accurate real-time image synthesis.
- Accelerated rendering throughput by introducing POSIX thread-based parallelism with a custom linear algebra engine and camera model, enabling multi-core utilisation and demonstrating expertise in low-level systems programming.

Scalable AI Photo Recognition & Clustering System *Personal Project C++, Qt/QML, OpenCV, Dlib, HNSWlib, QtConcurrent* [GitHub](#)

- Built a scalable AI-powered photo organisation system in C++ integrating OpenCV and Dlib, enabling automatic face detection and clustering across libraries of 10,000+ images with sub-second retrieval latency.
- Reduced face retrieval time to sub-linear complexity by implementing HNSW approximate nearest-neighbour search over 128-dimensional facial embeddings with persistent indexing, replacing brute-force linear scan.
- Improved application responsiveness by designing a fault-tolerant multithreaded image processing pipeline with asynchronous filesystem scanning and real-time UI updates, enabling smooth processing of large photo libraries without UI blocking.

WORK EXPERIENCE

Delivery Data Scientist *Jun 2025 – Jun 2026*
Turing (Google Gemini & related LLM projects) LLM Evaluation, Prompt Engineering, RLHF, Multimodal AI

- Contributed to 40+ AI training and evaluation projects for Google Gemini and related LLMs by curating, validating, and refining large-scale educational datasets spanning Mathematics, Physics, SAT, and GRE, directly improving training corpora quality.

- Improved model alignment by evaluating ground-truth responses across competing LLMs, identifying reasoning, factual, and LaTeX generation errors across text, image, and PDF modalities in a fast-paced agile workflow.
- Authored high-quality instructional datasets including SAT, GRE, JEE, and Physics questions with detailed solutions, hints, and rationales, supporting supervised fine-tuning and instruction-tuning pipelines at scale.
- Enhanced multimodal AI capabilities by creating and validating image-text datasets, generating educational visual content, and reviewing vision-language model outputs under strict quality standards across 40+ project cycles.
- Collaborated cross-functionally with engineering and research teams across rapidly evolving agile workflows involving preference ranking, synthetic data generation, and model evaluation.

AI Model Evaluator (Contract)*May 2025 – Jun 2026*
Outlier (LLM evaluation, C++/DSA focus) LLM Evaluation, Mathematical Reasoning, C++, Data Structures & Algorithms

- Evaluated and ranked outputs from multiple large language models on mathematics and C++/DSA problems, assessing correctness, algorithmic complexity, and instruction adherence to deliver structured preference rankings.
- Identified and documented logical, mathematical, and implementation errors through rigorous side-by-side model comparison, providing detailed justifications that guided model preference ranking and evaluation decisions.
- Validated AI-generated solutions for complex algorithmic problems, reviewing object-oriented design quality, computational complexity, and solution completeness while adapting quickly to evolving project specifications.

EDUCATION

IIT Kharagpur M.Sc. in Physics CGPA: 7.57/10	Kharagpur, India 2023 – 2025
Utkal University B.Sc. in Physics CGPA: 8.26/10	Odisha, India 2020 – 2023